

1. F, T, T, T, T
2. See image
 - a. $\lambda^3 - 5\lambda^2 + 8\lambda - 4$
 - b. Eigenvalues: 1, 2, 2
 - c. $E_1 = \text{span}([1 \ 1 \ 0]^T)$ $E_2 = \text{span}([1 \ 1 \ 0]^T)$
3.
 - a. $[0 \ 1]^T$
 - b. Yes, orthogonal vectors are linearly independent
 - c. No, the geometric multiplicity is not equal to the algebraic multiplicity
4.
 - a. No, the geometric multiplicity is not equal to the algebraic multiplicity
 - b. Eigenvalues: 4, -2, 2
 - c, d, e see image

$$1. \begin{array}{c|c} a & T \\ b & T \\ c & T \\ d & T \\ e & T \end{array}$$

$$2. \begin{array}{c|c} \cancel{A} & \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix} \end{array} \quad \begin{array}{l} \lambda \text{ is an eigenvalue of } \\ A \text{ iff } A\vec{v} = \lambda\vec{v} \text{ for} \\ \text{some non zero } \vec{v} \end{array}$$

$$n \text{ would be } 3 \quad \begin{array}{l} \vec{0} = \lambda I_n \vec{v} - A\vec{v} \\ \vec{0} = (\lambda I_3 - A)\vec{v} \end{array}$$

$$\downarrow \det(\lambda I_3 - A) = 0$$

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \lambda-1 & -4 & -5 \\ 0 & \lambda-2 & -6 \\ 0 & 0 & \lambda-2 \end{bmatrix}$$

$$(\lambda-1)(\lambda-2)(\lambda-2) + 0 + 0 - 0 - 0 - 0$$

b) Characteristic polynomial

$$(\lambda-1)(\lambda^2-4\lambda+4)$$

$$\lambda^3 - 4\lambda^2 + 4\lambda - \lambda^3 + 4\lambda - 4$$

$$= \lambda^3 - 5\lambda^2 + 8\lambda - 4$$

Eigenvalues: 1, 2, 2

$$c) \begin{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix} \end{pmatrix} N = N \begin{pmatrix} 0 & -4 & -5 \\ 0 & -1 & -6 \\ 0 & 0 & -1 \end{pmatrix}$$

2. c)

$$\begin{bmatrix} 0 & -4 & -5 \\ 0 & -1 & -6 \\ 0 & 0 & -1 \end{bmatrix} \vec{v} = \vec{0}$$

RREF

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow v_1 + v_2 + 0 \cdot v_3 = 0$$

$$E_1 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$E_2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -4 & -5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & -5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix} \cdot \vec{v} = \vec{0}$$

RREF

$$\begin{bmatrix} 1 & -4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow E_2 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)$$

3. a) $A = \begin{bmatrix} 8 & 4 \\ 0 & 8 \end{bmatrix}$

$E_8 = \cancel{N(A - 8I)} N\left(\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} - \begin{bmatrix} 8 & 4 \\ 0 & 8 \end{bmatrix}\right)$

$= N\left(\begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix}\right) \text{ RREF} \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot \vec{v} = \vec{0}$

$E_8 = \text{span}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

b.) yes, orthogonal vectors are linearly independent

c.) No, the geometric multiplicity is not equal to the algebraic multiplicity

4. a) No, the geometric multiplicity does not equal the algebraic multiplicity

b) eigenvalues: ~~4, -2, 2~~ 4, -2, 2

$$C) N \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 2 \\ 2 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -2 \\ -1 & 0 & -2 \\ -2 & -2 & -1 \end{pmatrix} N$$

$$\text{RREF: } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} E_4 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$$E_2 \left(\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 2 \\ 2 & 2 & 2 \end{pmatrix} \right) N = N \begin{pmatrix} 1 & -1 & -2 \\ -1 & 1 & -2 \\ -2 & -2 & 0 \end{pmatrix}$$

$$\text{RREF: } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} E_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

$$\det \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 2 \\ 2 & 2 & 2 \end{pmatrix} = -16 \quad A^{-1} = \begin{bmatrix} \frac{1}{8} & -\frac{3}{8} & \frac{1}{4} \\ \frac{3}{8} & \frac{1}{8} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix}$$

$$E_{-2} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$$

$$4) \quad A - 2I_3 = \begin{bmatrix} -1 & -1 & 2 \\ -1 & -1 & 2 \\ 2 & 2 & 0 \end{bmatrix} \quad \text{RREF} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} X_1 &= X_2 \\ X_3 &= 0 \end{aligned} \quad \vec{V} = X_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$A - I_3 = \begin{bmatrix} 0 & -1 & 2 \\ -1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad \text{RREF:} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X_1 = X_2 = X_3 = 0 \quad \vec{V} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{mult 2.}$$

So...

$$A = PDP^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad ?$$

could not invert

$$P = \begin{bmatrix} -1 & 1 & 1 \\ -1 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{3} \end{bmatrix}$$

4d) cont.

$$A = PDP^{-1} = \begin{bmatrix} -1 & 1 & 1 \\ -1 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{3} \end{bmatrix}$$

$$4e) A = PDP^T = \begin{bmatrix} -1 & 1 & 1 \\ -1 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

5 Let A be an $n \times n$ matrix with $\lambda = d$
 $N = \{ \vec{v} \in \mathbb{R}^n : A\vec{v} = d\vec{v}, \vec{v} \neq \vec{0} \}$

if V is a subspace of $\mathbb{R}^n$ then $\vec{0} \in V$

$V \neq \emptyset$

Let $\alpha, \beta \in \mathbb{R}$ and $\vec{u}, \vec{v} \in V$

$$\begin{aligned} A(\alpha \vec{u} + \beta \vec{v}) &= d(\alpha \vec{u} + \beta \vec{v}) \\ &= \alpha d \vec{u} + \beta d \vec{v} \\ &= \alpha A \vec{u} + \beta A \vec{v} \end{aligned}$$

$\therefore \alpha \vec{u} + \beta \vec{v} \in V \rightarrow V$ is closed

$\therefore N$ is a subspace of $\mathbb{R}^n$